

PAPER_1582 — Blood Glucose Level = [Glu] = SO_5·SO_5 = 100 mg/dL (cross-domain to Kármán line & MAD reciprocal!)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Biology **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209BB_S600 (Biology Unified Proof Set) lists **Blood Glucose Level** as an EXACT-tier UQFF closure:

[Glu] = SO_5·SO_5 = 100 mg/dL (cross-domain to Kármán line & MAD reciprocal!)

UQFF Closed Identity

[Glu] = SO_5·SO_5 = 100 mg/dL (cross-domain to Kármán line & MAD reciprocal!) EXACT
Pure integer-primitive expression with zero fitted constants.

Third Cross-Domain Reuse of SO_5²

SO_5² = 100 now governs **three unrelated physical domains**:

Observable	Domain	UQFF identity
Kármán line altitude	Space-physics	SO_5 ² = 100 km
MAD efficiency reciprocal	Black-hole accretion	1/SO_5 ² = 0.01
Blood glucose level	Human metabolism	SO_5 ² = 100 mg/dL

The same integer-primitive squared term sets: - The atmosphere/space boundary altitude - The Poynting flux conversion efficiency in magnetically-arrested black-hole disks - The healthy fasting glucose concentration in human blood
Different scales (km, dimensionless, mg/dL), same structural form. SO_5 = 10 is a universal numerical primitive across physical reality.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.
Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER 1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF’s predictive economy.

NOT REPLACEMENT

Empirical biology measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209BB_S600
- Related: PAPER_1494-1575 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "blood_glucose_100_mg_dl"})`

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